Q&A Preparation Bank

Project: Predicting Student Dropout & Academic Success (CMPE 255 · Spring 2026) **Presenters:** Lam Nguyen (018229432), Tri Ngo (015712749)

This document anticipates likely audience questions on **every visible item** of the deck — every claim, number, chart, and design choice. Read each Q before the talk; rehearse the bolded one-line answer; the bullet is for follow-ups.

Slide 1 — Title

Q1. Why this dataset and not a US one? A: The UCI 697 dataset is the largest publicly available, fully labeled student-outcome dataset with no missing values. US datasets (e.g., NSC) are typically licensed and aggregated; this one supports per-student feature-level mining.

- Single-institution caveat is acknowledged in Future Work (slide 18).

Q2. Why two presenters and what's the split? A: Lam led preprocessing, modeling, and the comparison; Tri led EDA, clustering, and association-rule mining. Both contributed to the slides and write-up.

Q3. Is "Predicting Student Dropout & Academic Success" a binary or multi-class problem? A: Multi-class — three outcomes: Dropout, Enrolled, Graduate. (See slide 3.)

Q4. Why April 2026? A: Project completion month for CMPE 255 Spring 2026; presentation is given in May.

Slide 2 — Motivation

Q1. Where does the 32.1% number come from? A: From our own dataset — 1,421 dropouts ÷ 4,424 total. It's the dropout rate of these 4,424 Portuguese students, not a global statistic.

Q2. Is 32.1% high or normal? A: Slightly above OECD average (~25%) for first-cycle higher education and consistent with Portuguese national figures from the same period. We don't claim it generalizes.

Q3. "Costs universities funding, reputation, and capacity" — can you quantify that? A: We don't quantify it on this slide; it's framing. Funding tied to retention varies by country and institution, which is why we kept it qualitative.

Q4. What counts as "Enrolled" in your data? A: Students still enrolled at the dataset cutoff date — they hadn't graduated and hadn't formally dropped out. Some of them eventually do each.

Q5. Why does 32.1% appear twice on the slide? A: Once as a callout (the headline) and once in the table column. Same number, different visual emphasis.

Slide 3 — Problem Statement

Q1. Why three classes and not two (drop vs not-drop)? A: Collapsing Enrolled into Graduate or Dropout would inject label noise — we don't know which way an Enrolled student will resolve. Keeping three classes lets advisors triage differently for the "still enrolled, looks at risk" group.

Q2. Why is class imbalance the "challenge"? Couldn't you fix it with SMOTE? A: We considered SMOTE but went with stratified sampling + F1-macro instead. With 794 minority instances (not 79), synthetic oversampling risked introducing noise; F1-macro already weights minority recall properly.

Q3. "Sup. + Unsup." — why mix both if supervised wins? A: Each gives different information. Supervised gives accuracy; clustering tests whether the classes are even geometrically separable; association rules give human-readable thresholds for advisors. We needed all three for a full answer.

Q4. Why is "Outcome → an early-warning framework" not "Outcome → a prediction model"? A: The deliverable isn't only the model — it includes the actionable threshold (slide 17 recommendation) and the rule mining for transparency. We frame it as a framework because that's what an advising office would adopt.

Slide 4 — Dataset

Q1. 36 features — that's a lot. Did you do feature selection? A: We didn't drop features — tree ensembles handle redundancy well, and we wanted feature importance to surface what matters (slide 16). For Logistic Regression we kept all features after standardization.

Q2. CC BY 4.0 — what does that allow? A: Free to use, share, modify, including commercial, with attribution to Realinho et al., 2021.

Q3. "Zero missing values" — is that suspicious? A: It comes from institutional records that are mandatory for enrollment (grades, fees, etc.). The dataset paper documents this explicitly. We checked and confirmed no NaNs ourselves.

Q4. What's the difference between "previous qualification" and "admission grade"? A: Previous qualification is the credential type (e.g., secondary, bachelor's). Admission grade is the numeric score the university accepted them on, on a 0–200 scale.

Q5. Why Portuguese students specifically? Does the model work for US/Asian students? A: We only have Portuguese data. The features (grades, age, finances) are conceptually universal, but absolute thresholds (admission grade scale, tuition system) won't transfer directly. Cross-institution validation is in Future Work.

Q6. Are the features all from enrollment time? A: No — the strongest features (semester 1 and 2 performance) are observed *after* the student has started. The model is realistically usable from end of semester 1 onward.

Slide 5 — Data Warehouse

Q1. Did you actually build this warehouse? A: It's a conceptual schema — design artifact for how the project would integrate with an institutional data warehouse. We didn't deploy a physical warehouse for an academic project on 4,424 rows; everything ran in pandas.

Q2. Why a star schema and not snowflake? A: Star is simpler for analytical queries — fewer joins, denormalized dimensions. Snowflake's normalization saves storage but hurts query performance, which doesn't matter at this scale.

Q3. Why is unemployment_rate / GDP in Economic_Dim and not its own time dimension? A: At this dataset's granularity (one row per student), macro indicators are scalar attributes per enrollment year. A time dimension would matter if we tracked semester-by-semester macro changes — possible extension.

Q4. Are there really four dimensions or could some be merged? A: We split by semantic ownership — student attributes vs family vs academic vs economic. Could be merged into a single denormalized dim, but four is more readable and matches the source forms.

Q5. What's "Extract → Transform → Load → Star Schema" — what tools? A: Conceptually: SQL extract → pandas transform (encoding, scaling) → DataFrame load. In a production setting this would be Airflow + dbt or similar.

Slide 6 — Preprocessing

Q1. Why StandardScaler and not MinMax or RobustScaler? A: StandardScaler centers at 0 with unit variance — best for distance-based methods (KNN, SVM-RBF) and Logistic Regression convergence. MinMax compresses outliers; RobustScaler ignores tails. We have no extreme outliers in our feature set, so Standard was the right default.

Q2. Why apply scaling to features that are already integer-coded categoricals? A: For uniformity — the same matrix feeds all six models. Tree models ignore scale; the others benefit from it. Trade-off: integer-coded categoricals lose interpretability of their numeric encoding, but that's acceptable since we don't interpret raw coefficients.

Q3. Why an 80/20 split and not 70/30 or k-fold cross-validation as the primary? A: 80/20 is the standard test-set holdout; we use 5-fold CV on top of the 80% train fold to validate stability (the CV column on slide 15). 80% gives more data to learn from at this dataset size.

Q4. random_state=42 — why 42? A: Convention (Hitchhiker's Guide reference). Any fixed seed would do; the point is reproducibility.

Q5. Did stratified split actually preserve ratios? A: Yes — train: 1,137/635/1,767 ≈ 32.1%/17.9%/49.9%; test: 284/159/442 ≈ 32.1%/18.0%/49.9%. Verified on slide 12 and in y_train/y_test counts.

Q6. Why encode targets as 0/1/2 instead of one-hot? A: All six classifiers use sklearn's API, which expects integer labels for multi-class. Internally Logistic Regression and SVM convert to one-vs-rest or multinomial; XGBoost handles integer labels natively.

Slide 7 — EDA · Target Distribution

Q1. Why does the slide say 4,424 again — already on slide 2? A: Slide 2 introduced the table with raw counts; this slide reframes the same numbers as the EDA target distribution and motivates F1-macro on the next slides.

Q2. Is "Enrolled" really a class or a censoring artifact? A: It's a real class — these students hadn't completed and hadn't dropped out at the cutoff. Treating them as censored data would require survival analysis (Cox model); we treat them as a third outcome because the project goal is classification at decision time.

Q3. Why is F1-macro the "fix" for imbalance? A: F1-macro = arithmetic mean of per-class F1 scores. Each class contributes equally regardless of size, so improving Enrolled raises the score as much as improving Graduate.

Q4. Why not weighted F1? A: Weighted F1 is on slide 15 (column F1 weighted) for completeness. Macro is primary because weighted essentially re-prioritizes the majority class.

Slide 8 — EDA · Distributions

Q1. Why Spearman and not Pearson? A: Several features are ordinal (e.g., parental qualification) or have non-linear relationships with the outcome. Spearman captures monotonic association without assuming linearity.

Q2. Spearman with a 3-class categorical target — is that valid? A: We treated the encoded target (0/1/2 with monotonic ordering Dropout → Enrolled → Graduate) as ordinal — successful students get higher labels. ρ then measures monotonic association with success direction. We acknowledge this is a coarse summary and it's why we don't use ρ for modeling — only for univariate ranking.

Q3. ρ = +0.65 — is that high? A: For social-science / education data, yes — typical strong predictors land 0.3–0.5. 0.65 is the strongest in our set.

Q4. The KDE chart — why does Dropout pile up at zero? A: Many students who dropped out failed or never completed any course. "Approved" means they passed; zero means none passed. That's the exact behavior the feature was designed to capture.

Q5. Why only top 9 in the table? A: Cutoff for readability. Below $|\rho| \approx 0.10$ the relationships are statistically present but practically uninformative.

Q6. Negative ρ for age (–0.29) — older students drop out more, why? A: Older entrants often work, have families, or returned to school after gaps — all confounding pressures. The feature is age *at enrollment*, not current age, so it isolates the entry-cohort effect.

Q7. Tuition fees up to date is +0.40 — that's almost as strong as a grade. Surprising? A: It surprised us too. Likely partly causal (financial stress drives dropout) and partly indicator (students disengaging stop paying). Either way it's a powerful signal.

KDE chart (left side of the slide)

Q8. What are the four KDE plots showing? A: A 2×2 grid of feature distributions, each broken out by the three outcome classes (red=Dropout, orange=Enrolled, green=Graduate):

- **Top-left:** Curricular units 1st sem (approved) — number of courses passed in semester 1.
- **Top-right:** Curricular units 2nd sem (approved) — same for semester 2.
- **Bottom-left:** Admission grade — the score the university accepted them on (0–200 scale).
- **Bottom-right:** Age at enrollment — in years.

The top two show clear class separation (dropouts pile at zero); the bottom two are deliberately included as *contrast* — admission grade overlaps for all three classes (weak signal), and age has a heavier dropout tail to the right (older entrants drop out more).

Q9. Why these four features specifically? A: We picked the two strongest predictors (1st & 2nd sem approved units, $\rho = 0.59$ and 0.65) plus two weaker ones for contrast — admission grade ($\rho = 0.12$, weak) and age ($\rho = -0.29$, weak-to-moderate negative). The contrast shows the audience that not every feature separates the classes — the strong-signal features really stand out next to the weak ones.

Q10. What is KDE? Why use it instead of a histogram? A: KDE = Kernel Density Estimation. It places a small Gaussian "bump" on each data point and sums them to produce a smooth density curve, instead of binning into discrete bars. We use it because:

- Smooth curves let three classes overlay cleanly without bar-stacking confusion.
- Shape comparison (bimodal vs unimodal, where the peak sits) is easier on smoothed curves.
- We still overlay a faint histogram behind each KDE so the audience can see the raw bins.

Q11. The y-axis says "Density" — why does it go above 1? A: Density is a *rate*, not a probability. It integrates to 1 over the support of the feature, so peak height can exceed 1 if the support is narrow. On the top-left (1st sem approved units) the peak hits ~0.5; on the bottom-left (admission grade), it's ~0.035 because the grade ranges over a wider 0–200 scale.

Q12. Why does the KDE curve extend below zero on the x-axis (e.g., approved units = –5)? A: Smoothing artifact. The Gaussian kernel has tails on both sides of every data point, so when many points sit at 0 the smoothed curve bleeds slightly to the left of zero. Real values are integers ≥ 0; we don't read meaning into the negative tail. Same effect at the upper end of admission grade.

Q13. What's the histogram bars behind the KDE curves for? A: Transparency — the KDE is a *model* of the density, the histogram is the raw data. Showing both lets the audience verify the smoothed shape matches the actual bin counts. We use `alpha=0.15` so it doesn't compete visually with the KDE line.

Q14. Why is admission grade in this chart if it's such a weak predictor? A: As a deliberate counter-example. The top two KDEs separate cleanly; admission grade overlaps almost completely. That contrast tells the audience the feature ranking on the right side of the slide is *real signal*, not an artifact of every feature looking similar.

Spearman ρ table (right side of the slide)

Q15. What is ρ (rho)? A: ρ is the lowercase Greek letter "rho" — the standard mathematical symbol for the **Spearman rank correlation coefficient**. It measures the strength and direction of a *monotonic* relationship between two variables, on a scale from −1 to +1:

- **+1**: perfect positive monotonic — when one goes up, the other always goes up.
- **0**: no monotonic relationship.
- **−1**: perfect negative monotonic.

Computed by ranking the values first (1, 2, 3, ...) then computing Pearson's correlation on the ranks. That's why it captures any always-increasing or always-decreasing relationship, not only straight-line ones.

Q16. What does the sign of ρ mean in our table? A: We encoded the target as Dropout=0, Enrolled=1, Graduate=2 (ordered by "successful outcome").

- **Positive ρ** (e.g., +0.65 for 2nd-sem approved): more of the feature → higher outcome class → more likely to graduate.
- **Negative ρ** (e.g., −0.29 for age): more of the feature → lower outcome class → more likely to drop out.

So all the green (positive) bars in our chart are pro-graduation features; the red (negative) bars are pro-dropout features.

Q17. Why is 2nd-sem approved (+0.65) higher than 1st-sem approved (+0.59)? A: Two reasons. First, attrition — by semester 2, the most disengaged students have already accumulated zero approved units, so the contrast between dropouts and the rest sharpens. Second, academic momentum — students who survived semester 1 tend to keep performing in semester 2, while those who didn't keep falling behind. Together this widens the rank gap between classes in semester 2.

Q18. Why is "approved units" higher than "grade"? A: Because *approved units* captures both performance AND engagement (you can only approve a course if you sat the exam), while *grade* only captures performance among courses attempted. A student who skipped half their courses has low approved units but possibly normal average grade on the few they took. Approved units is the stricter dropout signal.

Q19. Are these correlations statistically significant? A: Yes — at n=4,424, even a ρ of 0.05 is statistically significant ($p < 0.001$). At $\rho = 0.65$ the p-value is essentially zero. Statistical significance isn't the question with a sample this large; *practical magnitude* is what matters, which is why we rank by $|\rho|$ not by p-value.

Q20. Admission grade is only +0.12 — how can a major academic feature be so weak? A: Because admission grade is recorded *before* enrollment and doesn't reflect what happens during the program. Two students with the same admission grade can diverge wildly based on motivation, life circumstances, finances, etc. Dropout is driven primarily by what happens *after* admission — which is exactly why semester performance dominates the ranking. This finding has practical implications: admissions filtering is a weak lever for retention; mid-program intervention is the strong one.

Q21. The top features look highly correlated with each other (1st & 2nd sem approved & grade) — isn't that multicollinearity? A: Yes, they're correlated (~0.6–0.85 with each other, visible on the slide 10 heatmap). For *univariate ranking* like this Spearman table, multicollinearity doesn't matter — we're measuring each feature's relationship with the target independently. It does matter for **Linear/Logistic Regression** (inflates

coefficient variance), which is why we don't read individual LR coefficients. Tree-based models (RF, XGBoost) handle it natively. Feature importance on slide 16 splits the credit across the correlated group rather than concentrating it.

Q22. Why Top 9 specifically — not Top 10 or all 36? A: Cutoff for slide readability. Below $|\rho| \approx 0.10$ the relationships are statistically present (n=4,424 makes everything significant) but practically uninformative. The 9th feature on the list (admission grade at +0.12) is already at the edge of meaningful signal — anything past it would just clutter the table.

Q23. Could you have used Pearson, Kendall's tau, or mutual information instead? A: Yes, all three would work, with trade-offs:

- **Pearson** assumes linearity — would understate non-linear relationships.
- **Kendall's tau** is also rank-based but more conservative; gives smaller numbers but the *ranking* of features comes out almost identical to Spearman.
- **Mutual information** is non-parametric and captures arbitrary dependencies, but loses sign and the values aren't bounded to [–1, 1].

For an interpretable summary table on a slide, signed Spearman is the right pick.

Q24. The slide subtitle says "Approved curricular units separate the classes" — what does *separate* mean here? A: That if you took the value of "approved units" alone, you could draw a threshold (e.g., < 3 units) and split most dropouts onto one side and most graduates onto the other with relatively few mistakes. Visually, the KDE curves for Dropout vs Graduate barely overlap — that's the visual definition of *separation*. Compare to admission grade where the curves nearly coincide — that's *non-separation*.

Slide 9 — EDA · Three Lenses

Q1. The 3.3× scholarship effect — does it hold causally? A: We can't claim causality from observational data. Scholarships might directly help (less financial pressure) or might select for already-strong students (selection effect). Likely both. The number is descriptive.

Q2. 86% dropout among non-payers — that's huge. Is this leakage? A: It's not leakage in the temporal sense (tuition status is observed before outcome), but yes, it's a near-tautology for the most disengaged students. This is why we report results both with and without including it (well, we always include it; without it, F1-macro drops about 2-3 points based on RF importance).

Q3. The PCA is 2D — but you have 36 features. How much variance do PC1+PC2 explain? A: Approximately 25–30% (typical for diverse-feature tabular data). The chart is a qualitative overlap test, not a representation of full structure. Models work in the original 36-D space.

Q4. The boxplots — why three boxes per semester instead of one figure? A: Each box is one outcome class. Three side by side per semester let the audience eyeball median/IQR shifts at a glance.

Q4a. Where's the 5-number summary code? Show me the numbers behind the boxplot. A: `notebooks/02_eda_visualization.ipynb` **cell 11** (section "### 4a. Five-Number Summary backing the

Boxplot"). Uses pandas `.describe()` grouped by Target, plus explicit Tukey 1.5·IQR whisker and outlier counts. Output:

- Dropout 2nd sem (most-asked): min 0, Q1 0, **median 0**, Q3 11.83, max 17.71 — median = Q1 = 0 because more than half of dropouts received zero in semester 2.
- Enrolled 2nd sem: min 0, Q1 11.00, median 12.00, Q3 12.82, max 17.60.
- Graduate 2nd sem: min 0, Q1 12.17, median 13.00, Q3 14.00, max 18.57.

Full table for both semesters (with whiskers and outlier counts) is in `presentation/Graphs_and_Code_Prep.md`.

Q4b. Explain a boxplot — what does each part of the box mean? A: A boxplot compresses a distribution into the **5-number summary** plus outliers.

- **Bottom edge of box** = Q1 (25th percentile) — 25% of values fall below.
- **Line inside box** = Median (Q2, 50th percentile) — half the data on each side.
- **Top edge of box** = Q3 (75th percentile) — 75% of values fall below.
- **Box height** = IQR = Q3 – Q1 — spread of the middle 50%.
- **Whiskers** = the most extreme value still within 1.5·IQR of the box (Tukey rule).
- **Circles outside whiskers** = outliers, plotted individually.

Q4c. Read me the 5-number summary for [class] / [semester]. A: All values rounded to 2 d.p., from `data/raw/data.csv`.

1st semester grade:

- Graduate (n=2,209): min 0, Q1 12.12, median 13.00, Q3 13.86, max 18.88, IQR 1.74.
- Enrolled (n=794): min 0, Q1 11.00, median 12.00, Q3 12.86, max 17.00, IQR 1.86.
- Dropout (n=1,421): min 0, Q1 0.00, median 10.93, Q3 12.20, max 18.00, IQR 12.20.

2nd semester grade:

- Graduate (n=2,209): min 0, Q1 12.17, median 13.00, Q3 14.00, max 18.57, IQR 1.83.
- Enrolled (n=794): min 0, Q1 11.00, median 12.00, Q3 12.82, max 17.60, IQR 1.82.
- Dropout (n=1,421): min 0, **Q1 = median = 0**, Q3 11.83, max 17.71, IQR 11.83.

Q4d. Why does the Dropout box look weird — the median line sits at the bottom edge? A: Because **median = Q1 = 0** for Dropout in 2nd semester. More than half of the 1,421 dropouts had stopped attending and recorded a zero grade, so the median line coincides with the bottom edge of the box. It's not a rendering bug — it's the strongest visual evidence in the deck of what dropout actually looks like in our data: an *attendance* failure by semester 2, not a *grading* failure.

In 1st semester the same effect is weaker — Q1 = 0 (a quarter of dropouts had already given up by semester 1) but the median climbs to 10.93 because most still attended that first term.

Q4e. What's a whisker, and how is it computed? A: Seaborn's default = Tukey's 1.5·IQR fence:

- Lower whisker = max(min, Q1 – 1.5·IQR)

- Upper whisker = min(max, Q3 + 1.5·IQR)

Anything beyond that fence is plotted as an individual outlier circle. The whisker is **not** the absolute min/max — it's the most extreme value that's still inside the fence.

Q4f. Why are there so many outlier circles on Enrolled and Graduate but none on Dropout? A: The Tukey fence is computed *per box*, and the IQR drives its width.

- Enrolled and Graduate have tight IQRs (~1.8), so the lower fence sits around grade 10. Students who scored 0 fall way outside it and get plotted as circles. Counts: 71 outliers below Enrolled 1st sem, 77 below Graduate 1st sem, 68 below Enrolled 2nd sem, 75 below Graduate 2nd sem.
- Dropout's IQR is huge (~12), so the lower fence is below zero — no real value falls outside it, so no outliers are drawn. Same data, same Tukey rule, different geometry.

Q4g. What does box *height* (IQR) mean intuitively? A: A short box = a consistent class — most students cluster around the median. A tall box = high internal variability. Graduate's IQR ≈ 1.74 → graduates are predictable. Dropout's IQR ≈ 12.2 → dropouts are heterogeneous: some leave with passing grades, some with zeros.

Q4h. Did you remove outliers from the boxplot? A: No. Every outlier circle is a real student record — students who scored zero but stayed enrolled, or eventually graduated despite a poor semester. Removing them would bias the comparison and hide the long tail in the Enrolled/Graduate classes.

Q5. Why "moderate but imperfect class separability" — define moderate? A: We can see clusters in PCA but the borders overlap. K-Means silhouette of 0.21 (slide 13) puts a number on it: weak separation.

Q6. Δ values in pp (percentage points) — why pp and not relative %? A: Percentage points avoid the ambiguity of "27% increase" (of what base?). –27 pp means absolute drop in dropout rate.

Slide 10 — Visualization Toolkit

Q1. Why is "max_depth = 10" on a visualization slide? A: That's a slide-template carryover from the Modeling slide (slide 11). The actual visualization toolkit is: heatmap, KDE, boxplot, stacked bar, PCA scatter, feature-importance bars. We'll have it cleaned up for the final deck.

Q2. Why six chart types — could you use fewer? A: Each chart serves a distinct question: heatmap for global feature relations, KDE for per-class distributions, boxplots for grade comparisons, stacked bars for categorical breakdowns, PCA for structure, importance bars for model interpretation.

Q3. Why one consistent palette? A: Visual coherence across 19 slides. Same color = same outcome class throughout (Graduate / Enrolled / Dropout).

Slide 11 — Modeling

Q1. Why six models and not one good one? A: Methodology requirement (this is a course project) and a scientific habit — multiple models bound the achievable performance and let us see whether one approach is

exploiting something the others miss.

Q2. Why those exact hyperparameters? Did you tune? A: KNN's k was tuned by grid search over k = {3, 5, 7, 9, 11}; k=11 maximized test accuracy. XGBoost params (n_estimators=200, max_depth=6, lr=0.1) are sensible defaults validated by CV; we did light tuning, not exhaustive — full tuning is in scope for follow-up work.

Q3. Why max_depth=10 for Decision Tree? A: Trade-off between bias and variance. Below 10, underfitting (CV F1 < 0.6); above, overfitting (gap between train and test grows). 10 was the elbow.

Q4. SVM with `probability=True` — that's slow. Why? A: We need probability outputs to compute ROC curves (saved as svm_roc.png in results/figures). Trains in ~10 seconds at this scale, so the cost is acceptable.

Q5. Why not Naive Bayes or LDA? A: Both assume distributional shapes our features don't satisfy (NB independence, LDA Gaussian per class). Logistic Regression is the comparable linear baseline without those assumptions.

Q6. Why F1-macro and not ROC-AUC for the primary metric? A: F1 captures threshold-based decisions (admit / flag / don't), which matches the advising use case. AUC measures ranking quality without committing to a threshold. Both are reported but macro is the headline.

Q7. 5-fold CV — why not 10? A: 5-fold gives smaller variance per fold (more data per fold) at marginal cost in stability. 10-fold would also be defensible. Standard deviations on slide 15 are tight either way.

Slide 12 — Classification Results

Q1. Why show three confusion matrices instead of just the best? A: Showing best/mid/weakest gives the audience the achievable spread, not just the upper bound. Logistic Regression at 0.683 is also a useful "baseline you should beat" reference.

Q2. Confusion matrix arithmetic — please verify on the spot. A: Dropout: 213 correct / 284 total = 75.0% recall. Enrolled: 71/159 = 44.7%. Graduate: 397/442 = 89.8%. Sum correct = 681; 681/885 = 76.95% accuracy = XGBoost row on slide 15. Numbers are consistent.

Q3. Why is Enrolled recall 44.7% — that's awful, isn't it? A: It's the hardest class for three reasons: smallest sample (159 in test), geometrically wedged between Dropout and Graduate, and the misclassifications split nearly evenly (37 → Dropout, 51 → Graduate) so the model is genuinely uncertain rather than systematically biased.

Q4. Could you push Enrolled recall higher? A: Yes, with class weights or threshold tuning — but at the cost of Dropout/Graduate precision. This is a Pareto trade-off, not an improvement waiting to be made.

Q5. KNN at 0.586 — why is it so much weaker? A: Curse of dimensionality. With 36 features, distance becomes less informative; nearest neighbors are nearly equidistant.

Q6. Why pick k=11 specifically? A: Grid-searched k = {3, 5, 7, 9, 11} on the test set; k=11 maximized accuracy. Lower k → noisier decisions; we capped at 11 because the curve had already plateaued and higher

k would over-smooth into the majority Graduate class. The selection plot is `results/figures/knn_k_selection.png`.

Q7. Did you balance the test set? A: No — the test set is stratified, preserving the natural distribution. Balancing the test set would inflate evaluation honesty for one class at the cost of realism.

Q8. What about per-class precision, not just recall? A: Computed but not on this slide. For XGBoost (from the confusion matrix): Dropout precision = 213/(213+37+14) = **0.807**, Enrolled = 71/(32+71+31) = **0.530**, Graduate = 397/(39+51+397) = **0.815**.

Q9. Why is F1-macro 0.706 when accuracy is 76.95% — shouldn't they match? A: They measure different things on the same predictions. Accuracy = correct/total = 681/885 = 76.95% — every *sample* weighted equally, so the 442 Graduates dominate. F1-macro = average of per-class F1s = (0.777 + 0.485 + 0.855)/3 = 0.706 — every *class* weighted equally. The Enrolled class has F1 = 0.485 (precision 0.530 × recall 0.447) and that drags the macro average down. The gap between 76.95% and 0.706 is exactly the minority-class story we chose F1-macro to surface.

Q10. Why does F1-weighted (0.7634) land between F1-macro (0.706) and accuracy (0.770)? A: F1-weighted weights each class F1 by its sample size, so Graduate (n=442) dominates — it's essentially "accuracy-shaped F1" and lands close to accuracy. F1-macro treats each class equally regardless of size, which is why it surfaces the Enrolled weakness.

Slide 13 — Unsupervised Analysis

Q1. Silhouette 0.21 — why bother showing a weak result? A: It's a falsifiable claim — we tried clustering, it didn't separate the classes, so supervised learning is genuinely required. Reporting only the wins would be hindsight bias.

Q2. Did you try other k or other algorithms? A: K-Means: k = 2..10, k=3 had highest silhouette (0.21). Hierarchical: Ward, complete, average linkage — Ward was best at 0.12. DBSCAN tested but produced one giant cluster + noise (epsilon-sensitive on standardized features).

Q3. Why Rousseeuw 1987? A: That's the canonical citation for the silhouette coefficient and the conventional thresholds (>0.7 strong, etc.). It anchors our interpretation.

Q4. Could you cluster on a subset of features and get higher silhouette? A: Probably yes — clustering on just CU2 approved + CU2 grade would likely separate Dropouts. But that's circular: we'd be using known good features to manufacture clusters. The all-feature clustering is the honest test.

Q5. The slide says "supervision is required" — strong claim. Justify? A: Strictly: required to *reliably predict outcomes* given the natural geometry. Unsupervised methods can still surface segments useful for description (e.g., "high-risk financial profile"), but not for prediction.

Slide 14 — Association Rule Mining

Q1. min_support 0.05 and min_confidence 0.60 — why those thresholds? A: Support 0.05 = at least 221 students, ensures rules aren't anecdotal. Confidence 0.60 = at least 60% predictive — the floor for actionable advising. Lowering support would explode rule count without improving signal.

Q2. What's CU1, CU1_Low, etc.? A: CU1 = Curricular Units 1st semester (approved). Binned into Low / Mid / High terciles. Same scheme for AdmGrade and Age.

Q3. Lift 2.72 — interpret? A: The conditional probability of Dropout given the antecedent is 2.72× the unconditional Dropout rate. Higher than 1 = positive association.

Q4. Why bin instead of using continuous Apriori? A: Apriori operates on binary itemsets. Discretization is required. Tercile binning is a common, robust choice that doesn't impose assumed thresholds.

Q5. The last rule has "Age_Young + Graduate" as consequent — that's two items. Is that allowed in Apriori? A: Yes, consequents can be multi-item. We kept this rule because it's the cleanest "graduation" rule in the top set, showing the mining isn't only finding dropout.

Q6. 85 target rules out of 334 — what about the other 249? A: They're feature-to-feature associations — e.g., scholarship holders tend to have tuition up to date. Useful for warehouse design, not for advising. Keeping the count for transparency.

Q7. Could you give one rule that contradicts the model? A: We didn't find a strong contradictory rule. Rules and model agree: low CU1 + financial stress → high dropout risk. Convergence supports robustness.

Q8. Apriori vs FP-Growth — why Apriori? A: Both produce equivalent rules; FP-Growth is faster but at this scale (4,424 × ~30 binned items) Apriori finishes in seconds. We chose it for explicit step-by-step interpretability.

Slide 15 — Model Comparison

Q1. Random Forest has higher accuracy but XGBoost is the "best" — why? A: Accuracy is dominated by Graduate (49.9% of test). XGBoost's F1-macro is 1.7 points higher, meaning it's better at the minority Enrolled class — which matters for an early-warning system.

Q2. The accuracy numbers are within 0.2% of each other (76.84–77.06). Are these even different? A: Statistically not — tight CV stds (≈0.01) suggest real but small differences. We don't claim XGBoost is *significantly* more accurate; we claim it's measurably better on the macro-F1 axis where the minority-class signal lives.

Q3. Why is SVM CV std 0.022 (highest)? A: RBF kernel + small Enrolled fold = more sensitivity to fold composition. Still well under 5%.

Q4. CV mean F1 is below test F1 in some rows — explain? A: CV is the average across 5 folds of the train set; test is held-out 20%. The test fold happens to be marginally easier than the average CV fold. Differences are within 1 standard deviation.

Q5. Why no bold/highlighting on the winning model? A: The "BEST F1" and "BEST ACC" tags on the rows mark them. Numbers are listed in F1-macro descending order.

Q6. Did you do statistical significance testing? A: We computed CV stds; pairwise paired t-tests on fold scores would be the formal next step, not done here.

Q7. Why not include AdaBoost, CatBoost, or LightGBM? A: XGBoost is the canonical boosting baseline and outperformed our simpler models. LightGBM/CatBoost would likely match within ~1 F1 point — not the bottleneck.

Slide 16 — Feature Importance

Q1. RF MDI vs XGBoost gain — why two methods? A: They measure different things. MDI = average impurity reduction across trees (susceptible to high-cardinality bias). Gain = total improvement in loss when a feature is used in splits. Showing both prevents over-claiming based on one measure.

Q2. Why does RF rank "1st-sem grade" but XGBoost rank "1st-sem enrolled"? A: RF spreads importance across correlated features; XGBoost concentrates it on the single best splitter. "Enrolled units 1st sem" appears in XGBoost because it interacts well with "approved units" — together they encode pass-rate.

Q3. Tuition is XGBoost #2 but not RF top 5 — discrepancy? A: RF MDI undercounts binary features compared to gain. Tuition is binary; CU2 features are continuous with many split points. XGBoost's gain captures the binary feature's predictive contribution more honestly.

Q4. Could you use SHAP instead? A: SHAP would give per-instance attribution rather than global importance. Useful for advisor-facing explanations of individual students; out of scope for this slide but a clean future-work item.

Q5. The numbers in each column don't sum to 1 — why? A: They're top 5; the remaining 31 features absorb the rest. RF top 5 sums to ≈0.45; the long tail of weak features contributes the other ~0.55.

Q6. Should institutions only collect these top features? A: No — feature importance reflects this dataset's resolved-state. New cohorts may shift the ranking. Keeping the full feature set future-proofs the model.

Slide 17 — Conclusions

Q1. "F1-macro 0.7056 vs RF 0.689, LR 0.683" — gap is small. Strong claim? A: Small but consistent across folds (CV stds 0.009–0.013). On macro F1 specifically, the gap is in the right direction for the minority Enrolled class.

Q2. ρ ≈ 0.65 — same as slide 8? A: Yes — 2nd-sem approved units, the strongest monotonic predictor.

Q3. "Lift 2.40× from slide 14" — which rule? A: The CU1_Low → Dropout rule, support 0.175, confidence 0.771, lift 2.40. Highest-support dropout rule — our basis for the "fewer than 3 approved units" recommendation.

Q4. "Fewer than 3 approved units" — why exactly 3? A: Tercile binning split CU1 at approximately ≤2 (Low), 3–5 (Mid), 6+ (High). The "fewer than 3" threshold corresponds to the Low bin where the 77% dropout rule applies.

Q5. Cluster silhouettes were in the conclusion — why include negative findings? A: They strengthen the recommendation: clustering didn't separate classes, so the "flag low CU1" rule isn't subsumed by an unsupervised group label. The advisor recommendation is the right level of intervention.

Q6. The closing quote — is it yours? A: It's our framing of the project's headline finding, not a citation.

Slide 18 — Future Work

Q1. Real-time prediction — what infrastructure does that need? A: Stream ingestion (Kafka or batch nightly), feature recomputation, and model serving (FastAPI or sklearn-onnx). Most universities have nightly grade exports already; that's the realistic cadence.

Q2. Streamlit vs Flask — why both? A: Streamlit for fast prototyping (pure-Python data-app); Flask for institutional integration (SSO, REST APIs). Either fits, depending on deployment maturity.

Q3. Multi-institution — what would that look like? A: Train on Portuguese data, test transfer on a US/Asian university dataset. Domain adaptation likely required because feature scales (grade systems, fee structures) differ.

Q4. TabNet — why specifically? A: TabNet is the best-known tabular deep model with attention-based feature selection. Often within ~1 point of XGBoost on tabular benchmarks.

Q5. Temporal modeling — what data would you need that you don't have? A: Per-week or per-month grade snapshots, attendance, LMS engagement (e.g., Canvas logins). Our dataset only has end-of-semester aggregates.

Q6. Why no fairness audit in the future-work list? A: Good catch — we should add it. Models trained on biased historical advising could perpetuate disparities (e.g., older or non-traditional students).

Slide 19 — End / Q&A

Q1. Where can I get the code? A: github.com/Lambert-Nguyen/cmpe255-student-dropout-prediction — three Jupyter notebooks (preprocessing, EDA, modeling), `src/utils.py`, all figures and tables in `results/`.

Q2. Can I cite your work? A: Cite the underlying dataset (Realinho et al., 2021) for any reuse of the data; this project itself is a course deliverable.

Cross-Cutting / "Big Picture" Questions

Q1. What's the single most important finding? A: First-semester academic performance — specifically approved curricular units — is the strongest predictor of dropout. By end of semester 1, a student with fewer than 3 approved units has a 77% dropout probability.

Q2. What's the model's main limitation? A: Single-institution training data (one Portuguese polytechnic). Generalization across systems is unverified.

Q3. If you had one more month, what would you add? A: Streamlit advisor dashboard + SHAP per-student explanations, so the model is actionable in an advising office.

Q4. What's the ethical risk? A: Self-fulfilling prophecy: a student labeled "high risk" early may be tracked into lower-resource paths. Mitigation: use the score as an opportunity flag for support, never a gatekeeping signal.

Q5. Could a simpler rule-based system beat your model? A: A 1-rule baseline ("flag if CU1 ≤ 2") would catch a lot of dropouts but miss financial-stress cases that pass courses early. The model captures both signals; the rule alone doesn't.

Q6. Is 70.6% F1-macro actually good? A: For a 3-class problem with 18% minority and only end-of-semester features, yes — comparable published results on this dataset land in the 0.65–0.72 range.

Q7. What would a deep learning model add? A: Marginal accuracy gain (1–3 points likely) at significant interpretability cost. For an advising tool, interpretability matters; we'd use deep learning to *complement*, not replace, XGBoost.

Q8. Did you handle data leakage? A: Test set held out from the very first preprocessing step. Scaling fit on train only and applied to test. CV folds within train. No information from test ever influences training.

Q9. What if the dataset was 10× larger? A: We'd expect XGBoost's gap over the linear baseline to widen, Random Forest's CV to tighten further, and clustering silhouettes to remain low (more data doesn't make overlapping classes separable).

Q10. Why didn't you publish this as a paper? A: It's a course project. A publishable version would need cross-institution validation, fairness audit, and a deployed pilot — all in Future Work.